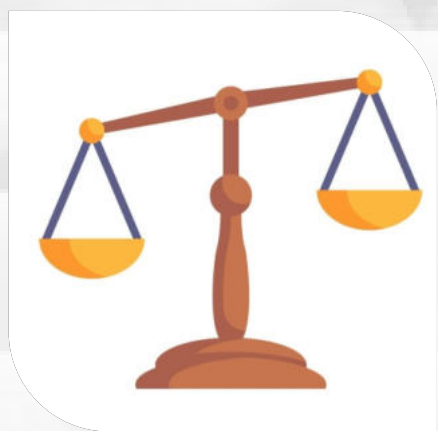


# UNIT 01

## PHYSICAL QUANTITIES AND MEASUREMENT



## INTRODUCTION OF PHYSICS

Physics is the study of the properties and interactions of matter and energy. It helps us understand how the universe works, from the smallest subatomic particles to the largest galaxy. It also helps us understand the fundamental laws that govern the behavior of matter and energy, such as Newton's laws of motion and the laws of thermodynamics.

### DEFINITION:

Physics is the branch of science that deals with the understanding and study of natural or physical phenomena involving the relation and interaction between matter and energy.

### FOR EXAMPLE:

- A ball rolling down a hill is an example of physics in action, as it involves the laws of motion and gravity.
- A car moving down the road is another example, as it involves the principles of energy, force, and motion.
- A light bulb turning on is another example, as it involves the principles of electricity and magnetism.

Physics is all around us and plays a crucial role in our everyday lives by helping us understand how things work and how we can harness and control the laws of Physics to improve our lives.

## BRANCHES OF PHYSICS

**1. Mechanics:** Mechanics is the branch of physics that deals with the study of motion and the forces that cause motion. It includes the study of objects at rest, objects in motion, and the forces that affect their motion.

### Example:

- ▶ A ball rolling down a hill.
- ▶ A car moving down the road.



**2. THERMODYNAMICS:** Thermodynamics is the branch of physics that deals with the study of heat and temperature and their relationship to energy and work. It includes the study of heat transfer, thermodynamic systems, and the laws of thermodynamics.

### Example:

- ▶ A car engine converting fuel into energy.
- ▶ A refrigerator keeping food cool.
- ▶ A steam engine converting heat into mechanical work.



**3. ELECTRICITY:** Electricity is the branch of physics that deals with the study of electric charge and electric fields. It includes the study of electric circuits, electric potential, and electromagnetic waves.

## Example:

- ▶ A light bulb turning on.
- ▶ A phone charging.
- ▶ A thunderstorm.



**4. MAGNETISM:** Magnetism is the branch of physics that deals with the study of magnets and magnetic fields. It includes the study of magnetic forces and magnetic materials.

## Example:

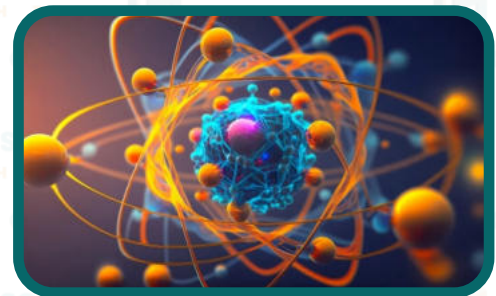
- ▶ A magnetic compass.
- ▶ A refrigerator magnet.
- ▶ An electric motor.



**5. ATOMIC PHYSICS:** Atomic Physics is the branch of physics that deals with the study of atoms and their structure. It includes the study of atomic spectra, the behavior of electrons in atoms, and the properties of ions and atomic nuclei.

## Example:

- ▶ How atoms emit light?
- ▶ Behavior of electrons in atoms.



**6. BIO-PHYSICS:** Bio-physics is the branch of physics that applies the principles of physics to the study of living organisms and their functions. It includes the study of biochemistry, cell biology, and genetics.

## Example:

- ▶ Study of how muscles contract.
- ▶ Behaviour of enzymes.
- ▶ Study of the nervous system.



**7. OPTICS:** Optics is the branch of physics that deals with the study of light and its properties. It includes the study of reflection, refraction, and the behavior of light through different materials.

## Example:

- ▶ How a mirror reflects light?
- ▶ How a rainbow is formed?
- ▶ How a camera works?



**8. SOUND:** Sound is the branch of physics that deals with the study of sound waves and their properties. It includes the study of sound production, sound transmission, and the behavior of sound in different media.

**Example:**

- ▶ How a guitar produces sound?
- ▶ How sound travels through air?
- ▶ How sound is used in sonar technology?

**9. PLASMA PHYSICS:** Plasma Physics is the branch of physics that deals with the study of plasma, a gas-like state of matter that consists of charged particles. It includes the study of the behavior of plasma, its interactions with magnetic fields, and its role in natural phenomena such as lightning and auroras.

**Example:**

- ▶ How the sun produces energy?
- ▶ How lightning occurs?
- ▶ How the Northern Lights are formed?

**10. GEOPHYSICS:** Geophysics is the branch of physics that deals with the study of the Earth and its physical properties. It includes the study of earthquakes, plate tectonics, and the Earth's magnetic field.

**Example:**

- ▶ How the sun produces energy?
- ▶ How lightning occurs?
- ▶ How the Northern Lights are formed?

**11. ASTROPHYSICS:** Astrophysics is a branch of physics that deals with the study of celestial objects and phenomena, such as stars, planets, galaxies, and black holes. It helps us understand the properties and behavior of these objects, as well as the origins and evolution of the universe.

**Example:**

- ▶ Study of star formation and evolution.
- ▶ Study of Planet.
- ▶ Study of Galaxies.

**12. PARTICLE PHYSICS:** Particle physics is a branch of physics that deals with the study of the smallest building blocks of matter, called particles. It helps us understand the properties and behavior of these particles, as well as the fundamental laws that govern their interactions.



**Example:**

- ▶ Study of subatomic particles such as protons, neutrons, and electrons. Nuclear Physics: Nuclear physics is the study of the structure and behavior of atomic nuclei. It is a branch of physics that deals with the properties and interactions of subatomic particles, such as protons and neutrons.



**13. NUCLEAR PHYSICS:** Nuclear physics is the study of the structure and behavior of atomic nuclei. It is a branch of physics that deals with the properties and interactions of subatomic particles, such as protons and neutrons.

**Example:**

- ▶ Nuclear power plants.
- ▶ Radiation therapy in medical treatment.

**PHYSICAL QUANTITIES**

All measurable quantities are called physical quantities such as length, mass, time and temperature. A physical quantity possesses at least two characteristics in common. One is its numerical magnitude and the other is the unit in which it is measured.

**TYPES OF PHYSICAL QUANTITIES**

Physical quantities are divided into two quantities:

1. Base or fundamental quantities
2. Derived quantities

**FUNDAMENTAL PHYSICAL QUANTITIES**

Fundamental physical quantities are basic measurements that are used to describe the physical properties of matter and energy. They are the building blocks of all other physical quantities, and their standard units are used as a reference for all other measurements.

**Example:**

Here are a few examples of fundamental physical quantities and their standard units:

1. **Length:** The measurement of the platinum-iridium distance between two points. The standard unit for length is the metre (**m**).
2. **Mass:** The measurement of the amount of matter in an object. The standard unit for mass is the kilogram.
3. **Time:** The measurement of the duration of an event. The standard unit for time is the second (**s**).

## FUNDAMENTAL QUANTITIES

| Physical Quantities | Symbol               | Units    | Symbol of Unit |
|---------------------|----------------------|----------|----------------|
| Length              | <b>l</b>             | Metre    | <b>m</b>       |
| Mass                | <b>m</b>             | Kilogram | <b>kg</b>      |
| Time                | <b>t</b>             | Second   | <b>s</b>       |
| Electric Current    | <b>I</b>             | Ampere   | <b>A</b>       |
| Temperature         | <b>T</b>             | Kelvin   | <b>K</b>       |
| Luminous Intensity  | <b>I<sub>v</sub></b> | Candela  | <b>cd</b>      |
| Amount of Substance | <b>n</b>             | Mole     | <b>mol</b>     |

## DERIVED QUANTITIES

Physical quantities which are explained on the basis of base physical quantities are called derived quantities.

**Example:**

**Speed:** Speed is a derived quantity because it can be calculated by dividing distance by time (**speed = distance/time**).

**Acceleration:** Acceleration is calculated by dividing the change in velocity by the change in time (**acceleration = change in speed/time**).

**Force:** Force is calculated by multiplying the mass by acceleration (**force = mass x acceleration**).

## DERIVED QUANTITIES

| Physical Quantities | Symbol   | Units                     | Symbol of Unit                |
|---------------------|----------|---------------------------|-------------------------------|
| Speed               | <b>v</b> | Metre/Second              | <b>m/s</b>                    |
| Acceleration        | <b>a</b> | Metre/Second <sup>2</sup> | <b>m/s<sup>2</sup></b>        |
| Volume              | <b>V</b> | Cubic Metre               | <b>m<sup>3</sup></b>          |
| Force               | <b>F</b> | Newton                    | <b>N (kg m/s<sup>2</sup>)</b> |
| Pressure            | <b>P</b> | Pascal                    | <b>Pa (N/m)</b>               |

|        |          |         |                |
|--------|----------|---------|----------------|
| Work   | <b>W</b> | Joule   | <b>J (N-m)</b> |
| Charge | <b>Q</b> | Coulomb | <b>C (A.s)</b> |

## MEASUREMENT & MEASURING INSTRUMENTS

Measurement is the process of determining the size, amount, or degree of something. In science and engineering, measurements are used to gather data and make calculations.

### UNIT

Measurements are usually expressed in units. Units are a way to indicate the amount of a quantity.

## MEASURING INSTRUMENTS

Measuring instruments are tools that are used to measure physical quantities such as length, mass, time, and temperature etc.

### Example:.

- ▶ **Ruler:** A ruler is used to measure length or distance.
- ▶ **Scale:** A scale is used to measure weight or mass.
- ▶ **Clock:** A clock is used to measure time.
- ▶ **Thermometer:** A thermometer is used to measure temperature.
- ▶ **Voltmeter:** A voltmeter is used to measure electric potential difference or voltage.
- ▶ **Ammeter:** An ammeter is used to measure electric current.

### S.I UNIT

The International System of Units (**SI**) is the standard system of units used for measurement in science and technology. It is based on seven base units:

## THE STANDARD OF LENGTH

The length is defined as the minimum distance between two points lying on the same plane. The S.I unit of length is meter (**m**).

### Length:

The length of the path traveled by light in vacuum during the time interval of  $1/299792458$  of a second.

### METER RULE

A meter rule is a length measuring instrument. It is commonly used in the laboratories to measure length of an object or distance between two points. It is one-metre-long which is equal to 100 centimeters. Each centimeter (**cm**) is divided into 10 small divisions called millimeter (**mm**). Thus one millimeter is the smallest reading that can be taken using a meter rule and is called its least count.



## WORKING

To use a metre rule, simply line up the starting point of the measurement with the "0" mark on the ruler, and read the measurement at the endpoint. It's important to make sure the ruler is straight and not bent when taking measurements.

## VERNIER CALIPER

The accuracy obtained in measurements using a meter rule is up to 1 mm. However, an accuracy greater than 1 mm can be obtained by using some other instrument such as a Vernier Calipers. A Vernier Caliper is a measuring tool that is used to measure internal and external distance extremely accurate. It consists of two jaws. One is a fixed jaw with main scale attached to it. Main scale has centimeter and millimeter marks on it. The other jaw is a moveable jaw. It has Vernier scale having 10 divisions over it such that each of its division is 0.9 mm. The difference between one small division on main scale division and one Vernier scale division is 0.1 mm. It is called least count (LC) of the Vernier calipers. Least count of the Vernier calipers can also be found as given below:



$$L.C = \frac{\text{Smallest reading on main scale}}{\text{no. of division on vernier scale}}$$

$$L.C = \frac{1\text{mm}}{10 \text{ division}}$$

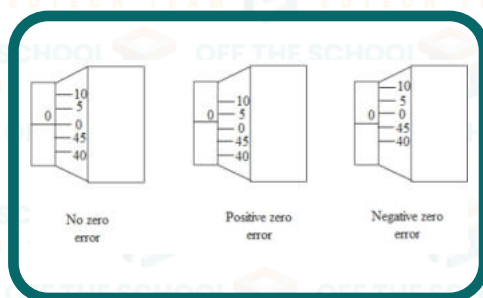
$$L.C = 0.1 \text{ mm} = 0.01\text{cm}$$

## WORKING OF A VERNIER CALIPERS

First of all, find the error, if any, in the measuring instrument. It is called the zero error of the instrument. Knowing the zero error, necessary correction can be made to find the correct measurement. Such a correction is called zero correction of the instrument.

## ZERO ERROR AND ZERO CORRECTION

In order to find the zero error, close the jaws of Vernier Calipers gently. If zero line of the Vernier scale coincides with the zero of the main scale, then the zero error is zero. If the zero error will exist if zero line of the Vernier scale is not coinciding with the zero of main scale. Zero error will be positive if the zero line of Vernier scale is on the right side of the zero of the main and will be negative if zero line of the Vernier scale is on the left side of zero of the main scale.



## MICROMETER SCREW GAUGE

A micrometer screw gauge is an instrument that is used to measure small lengths with accuracy greater than a Vernier callipers.



**WORKING**

It consists of a U-shaped metal frame with a metal stud at its one end. A hollow cylinder has a millimeter scale over it along a line called index line parallel to its axis. The hollow cylinder acts as nut. It is fixed at the end of U-shaped frame opposite to the stud. A Thimble has a threaded spindle inside it. As the thimble completes one rotation, the spindle moves 1 mm along the index line. It is because the distance between consecutive threads on the spindle is 1 mm. This distance is called the pitch of screw on the spindle. The thimble has 100 divisions around its one end. It is the circular scale of the screw gauge. The Least count of a screw gauge can be found as given below:

$$\text{L.C} = \frac{\text{pitch of the screw gauge}}{\text{no. of divisions on circular scale}}$$

$$\text{L.C} = \frac{1\text{mm}}{100}$$

$$\text{L.C} = 0.01 \text{ mm} = 0.001 \text{ cm}$$

**ZERO ERROR AND ZERO CORRECTION**

In order to find the zero error, close the gap between the spindle and the stud of the screw gauge by rotating the ratchet in the clockwise direction. If zero of circular scale coincides with the index line, then the zero error will be zero. Zero error will be positive if zero of circular scale is behind the index line. In this case, multiply the number of divisions of the circular scale that has not crossed the index line with the least count of screw gauge to find zero error. Zero error will be negative if zero of circular scale has crossed the index line. In this case, multiply the number of divisions of the circular scale that has crossed the index line with the least count of screw gauge to find the negative zero error.

**THE STANDARD OF MASS**

The kilogram is the S.I unit of mass and is equal to the mass of the international prototype of the kilogram, a platinum-iridium standard that is kept at the international Bureau of Weights and measures.

**KILOGRAM**

The kilogram is a cylinder of spherical metal about 39 millimeters wide by 39 millimeters tall that serves as the world's mass standard.

**PHYSICAL BALANCE**

A physical balance is a device used to measure the weight or mass of an object. It works by comparing the unknown weight to a known weight, called a standard weight.

**WORKING**

It consists of a beam with a pivot point in the middle and two pans hanging from either side. The unknown weight is placed on one pan, and standard weights are added to the other pan until the beam is level and the two pans are in balance.



**ELECTRONIC BALANCE**

An electronic balance is a type of scale that uses electronic sensors to measure the weight of an object. It typically displays the weight on a digital screen and can be used to weigh items with a high level of accuracy.

**THE STANDARD OF TIME**

The Second is the S.I unit of time.

**SECOND**

The second is defined as 9192631770 times the period of vibration of radiation from the cesium atom.

**STOPWATCH**

A stopwatch is used to measure the time interval of an event. There are two types of stopwatch: (i) Mechanical stopwatch (ii) Digital stopwatch

**MECHANICAL STOPWATCH**

A mechanical stopwatch can measure a time interval up to a minimum 0.1 second. A mechanical stopwatch has a knob that is used to wind the spring that powers the watch. It can also be used as a start-stop and reset button. The watch starts when the knob is pressed once. When pressed second time, it stops the watch while the third press brings the needle back to zero position.

**DIGITAL STOPWATCH**

The digital stopwatch starts to indicate the time lapsed as the start/stop button is pressed. As soon as start/stop button is pressed again, it stops and indicates the time interval recorded by it between start and stop of an event. A reset button restores its initial zero setting

**PREFIXES**

Prefixes are useful for expressing units of physical quantities that are either very big or very small.

**UNIT PREFIX**

A unit prefix is a specifier. It indicates multiplier or fractions of the units.

| Multiple  | Prefix      | Symbol   |
|-----------|-------------|----------|
| $10^{18}$ | <b>Exa</b>  | <b>E</b> |
| $10^{15}$ | <b>Peta</b> | <b>P</b> |

|            |              |       |
|------------|--------------|-------|
| $10^{12}$  | <b>Tera</b>  | T     |
| $10^9$     | <b>Giga</b>  | G     |
| $10^6$     | <b>Mega</b>  | M     |
| $10^3$     | <b>Kilo</b>  | k     |
| $10^2$     | <b>Hecto</b> | h     |
| $10^1$     | <b>Deca</b>  | da    |
| $10^{-1}$  | <b>Deci</b>  | d     |
| $10^{-2}$  | <b>Centi</b> | c     |
| $10^{-3}$  | <b>Milli</b> | m     |
| $10^{-6}$  | <b>Micro</b> | $\mu$ |
| $10^{-9}$  | <b>Nano</b>  | n     |
| $10^{-12}$ | <b>Pico</b>  | p     |
| $10^{-15}$ | <b>Femto</b> | f     |
| $10^{-18}$ | <b>Atto</b>  | a     |

### SCIENTIFIC NOTATION

The method of writing extremely large or small numbers in science is known as scientific notation. Numbers in scientific notation are made up of three parts. The coefficient, base and the exponent.

### RULES

There are a few rules followed in scientific notation:

The coefficient must be equal to or greater than one (Not Zero).

The base must be 10.

The exponent can be negative or positive.

**NOTE**

When performing calculations with numbers having different amounts of significant figures, it is essential to round the final answer to the same number of significant figures as the number with the least significant figures.

**SIGNIFICANT FIGURE**

Significant figures are the digits in a number that provide information about its precision. They are used to indicate the level of uncertainty in a measurement.

**RULES**

There are a few rules that need to be followed for digits to be significant: All non-zero digits are always significant. For example, the number 42 has two significant figures. Zeros between two non-zero digits are always significant. For example, the number 401 has three significant figures. Zeros on the right of the last non-zero digit and on the left of a decimal point are significant. For example, the number 4.01 has three significant figures. Zeros on the right of the last non-zero digit and on the right of a decimal point are not always significant. For example, the number 4.00 has only two significant figures.

**ACCURACY**

In physics, accuracy refers to how close a measurement is to the actual or accepted value.

**EXPLANATION**

In other words, it is a measure of how close a measurement is to the actual value of the quantity being measured. A measurement that is very accurate will be very close to the true value, while a measurement that is not very accurate will be farther away.

**VOLUME AND ITS MEASUREMENT**

For density to be measured or calculated we first need to find the volume of substances. Most solid shapes have formulae for their volume which is obtained through different parameters such as radius, height, depth, width, base and length, but for irregular objects, liquids and gases this approach is unusual. The volume of liquids can be measured with the help of Cylinders, and Beakers.

**MEASURING CYLINDER**

Measuring cylinder is a glass or plastic cylinder with a scale-graduated in cubic centimeters or millimeters (ml). It is used to find the volume of liquids. When a liquid is poured, it rises to a certain height in the cylinder. The level of the liquid in the cylinder is noted and volume of the liquid is obtained.

**REGULAR SOLID**

If an object has a regular shape its volume can be calculated. For instance:

Volume of a rectangular block = length  $\times$  width  $\times$  height

Volume of a cylinder =  $\pi \times \text{radius}^2 \times \text{height}$

**IRREGULAR SOLID**

For an irregular solid its volume is calculated by lowering the object in a partially filled measuring cylinder. The rise in the level on the volume scale gives the volume of that object. Thus the volume of irregular solid is calculated by subtracting the original volume of liquid from the raised volume of liquid of liquid.

**DENSITY AND ITS MEASUREMENT**

The term density of a substance is defined as mass of substance ( $m$ ) per unit volume ( $V$ ). It is denoted by Greek letter  $\rho$  (rho).

$$\rho = \frac{m}{V}$$

Density is characteristic property of any pure substance. Objects made of a particular pure substance such as pure Gold can have any size or mass but its density will be same for each. The S.I unit of density is  $\text{kg/m}^3$ .

**MEASURING THE DENSITY**

There are two ways of finding the density of a substance either mathematically or experimentally by taking density of water at  $4^\circ\text{C}$  as a reference which is sometimes known as relative density or specific density.

It has no unit.

It is a number whose value is the same as that of the density.

$$\text{Relative density} = \frac{\text{density of substance}}{\text{density of water}}$$